

SKILL – 6

2520090160 – Snigdha

```
#!/bin/bash
```

```
echo "----- ESCAPE SEQUENCES -----"
```

```
echo "Hello\nWorld"
```

```
echo -e "Hello\nWorld"
```

```
echo "Hello\tWorld"
```

```
echo -e "Hello\tWorld"
```

```
echo "He said \"Hello\""
```

```
echo 'It\'s Linux'
```

```
echo "----- ESCAPED SPACE -----"
```

```
echo Hello\ World
```

```
file=My\ File
```

```
echo "$file"
```

```
echo "----- SPECIAL SYMBOLS -----"
```

```
echo "Dollar: \$100"
```

```
echo "Hash: \#Linux"
```

```
echo "Backslash: \\"
```

echo "Ampersand: \&"

echo "Asterisk: *"

echo "----- COMPLEX INPUT -----"

echo "File name: My\ File"

echo "Price: \\$500"

echo "Path: C:\\Users\\Student"

echo "----- ESCAPE TEST COMPLETED -----"

```
snigdha@Snigdha:~/quotes_practical$ cd ~/quotes_practical
snigdha@Snigdha:~/quotes_practical$ nano escape.sh
snigdha@Snigdha:~/quotes_practical$ chmod +x escape.sh
snigdha@Snigdha:~/quotes_practical$ ./escape.sh
----- ESCAPE SEQUENCES -----
Hello\nWorld
Hello
World
Hello\tWorld
Hello  World
He said "Hello"
It's Linux
----- ESCAPED SPACE -----
Hello World
My File
----- SPECIAL SYMBOLS -----
Dollar: $100
Hash: \#Linux
Backslash: \
Ampersand: \&
Asterisk: \*
----- COMPLEX INPUT -----
File name: My\ File
Price: $500
Path: C:\Users\Student
----- ESCAPE TEST COMPLETED -----
snigdha@Snigdha:~/quotes_practical$ bash -n escape.sh
snigdha@Snigdha:~/quotes_practical$ |
```

#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

```
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("Parent Process PID: %d\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child Process PID: %d\n", getpid());
        printf("Child: Executing ls command...\n");

        execl("/bin/ls", "ls", "-l", NULL);

        perror("execl failed");
        exit(1);
    }
    else
    {

```

```

printf("Parent: Waiting for child...\n");

wait(NULL);

printf("Parent: Child process completed.\n");
}

return 0;
}

```

```

snigdha@Snigdha:~/quotes_practical$ nano process.c
snigdha@Snigdha:~/quotes_practical$ gcc -Wall -Wextra -o process process.c
snigdha@Snigdha:~/quotes_practical$ ./process
Parent Process PID: 903
Parent: Waiting for child...
Child Process PID: 904
Child: Executing ls command...
total 28
-rwxr-xr-x 1 snigdha snigdha 562 Sep 30 03:56 escape.sh
-rwxr-xr-x 1 snigdha snigdha 16248 Sep 30 03:59 process
-rw-r--r-- 1 snigdha snigdha 654 Sep 30 03:59 process.c
-rwxr-xr-x 1 snigdha snigdha 602 Sep 30 03:49 quotes.sh
Parent: Child process completed.
snigdha@Snigdha:~/quotes_practical$ ls -l
total 28
-rwxr-xr-x 1 snigdha snigdha 562 Sep 30 03:56 escape.sh
-rwxr-xr-x 1 snigdha snigdha 16248 Sep 30 03:59 process
-rw-r--r-- 1 snigdha snigdha 654 Sep 30 03:59 process.c
-rwxr-xr-x 1 snigdha snigdha 602 Sep 30 03:49 quotes.sh
snigdha@Snigdha:~/quotes_practical$ |

```

```

#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

```

```

int main()
{
    printf("Trying to execute a program...\n");
}

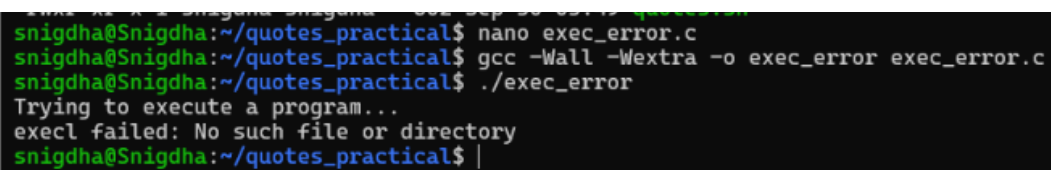
```

```
execl("/bin/nonexistent", "nonexistent", NULL);
```

```
perror("execl failed");
```

```
return 1;
```

```
}
```



```
snigdha@Snigdha:~/quotes_practical$ nano exec_error.c
snigdha@Snigdha:~/quotes_practical$ gcc -Wall -Wextra -o exec_error exec_error.c
snigdha@Snigdha:~/quotes_practical$ ./exec_error
Trying to execute a program...
execl failed: No such file or directory
snigdha@Snigdha:~/quotes_practical$ |
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
int main()
```

```
{
```

```
    pid_t pid = fork();
```

```
    if (pid < 0)
```

```
    {
```

```
        perror("fork");
```

```
        return 1;
```

```
    }
```

```
    if (pid == 0)
```

```

{
    printf("Child: Running echo with arguments\n");

    execl("/bin/echo",
        "echo",
        "Hello",
        "from",
        "Child",
        "Process",
        NULL);

    perror("execl");
    exit(1);
}

else
{
    wait(NULL);

    printf("Parent: Child finished.\n");
}

return 0;
}

```

```

snigdha@Snigdha:~/quotes_practical$ nano arguments.c
snigdha@Snigdha:~/quotes_practical$ gcc -Wall -Wextra -o arguments arguments.c
snigdha@Snigdha:~/quotes_practical$ ./arguments
Child: Running echo with arguments
Hello from Child Process
Parent: Child finished.

```